CHAPTER 3 / SUPPLEMENTARY MATERIAL

Cyber–Physical Security and Mission Assurance for Unmanned Aerial Vehicles

Threat Modeling and Security Assessment Methodologies for UAS

Learning goal: connect system interfaces to threats, controls, and verifiable evidence.

Threat-modeling fundamentals

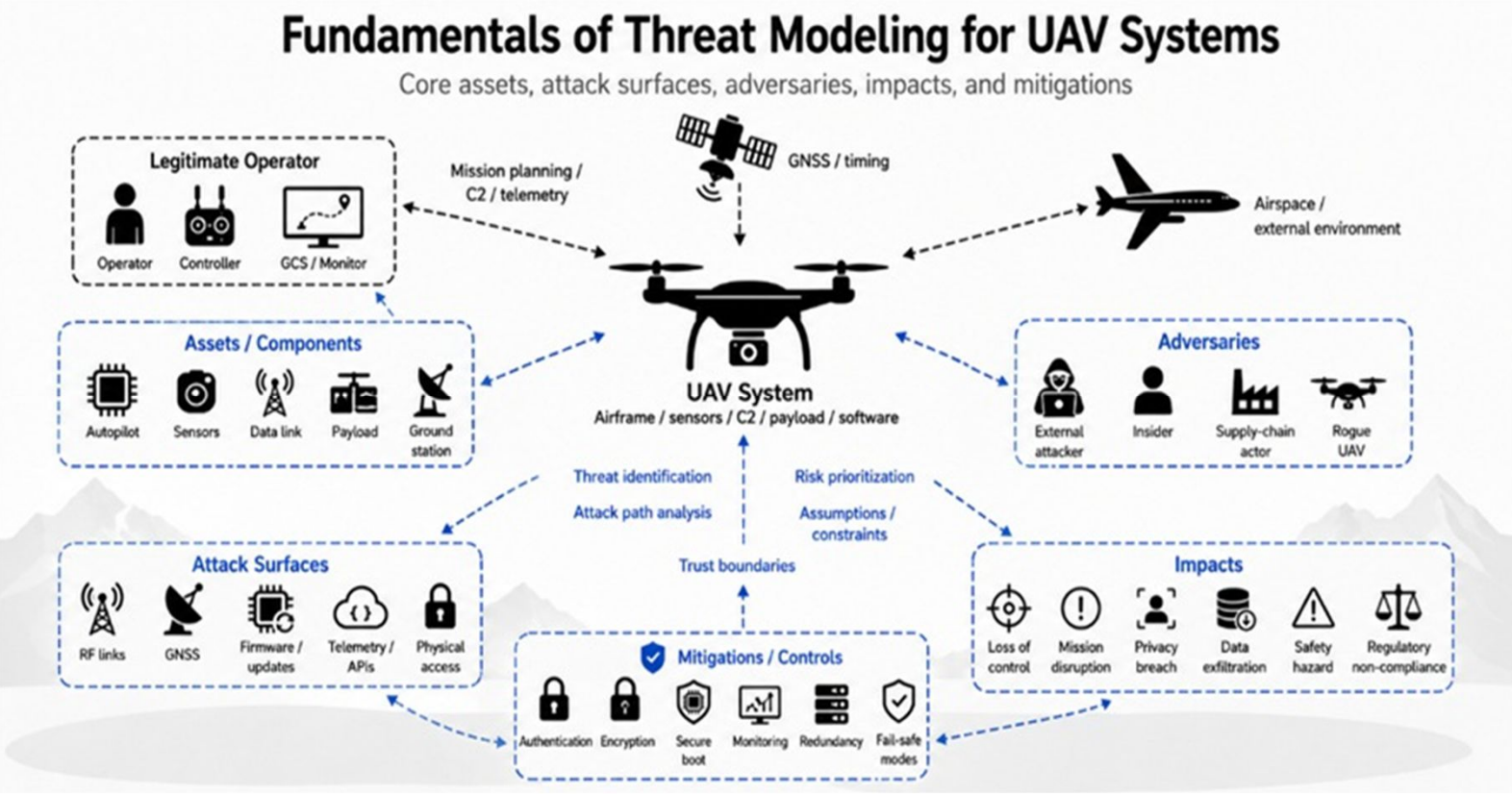


Figure S3.1. A threat model relates assets, exposed interfaces, adversary assumptions, mission consequences, and controls. Its scope determines what the assessment can claim.

Conceptual relationships, not a demonstrated attack sequence. Teaching notes expand the assessment questions.

Attack surfaces and trust boundaries

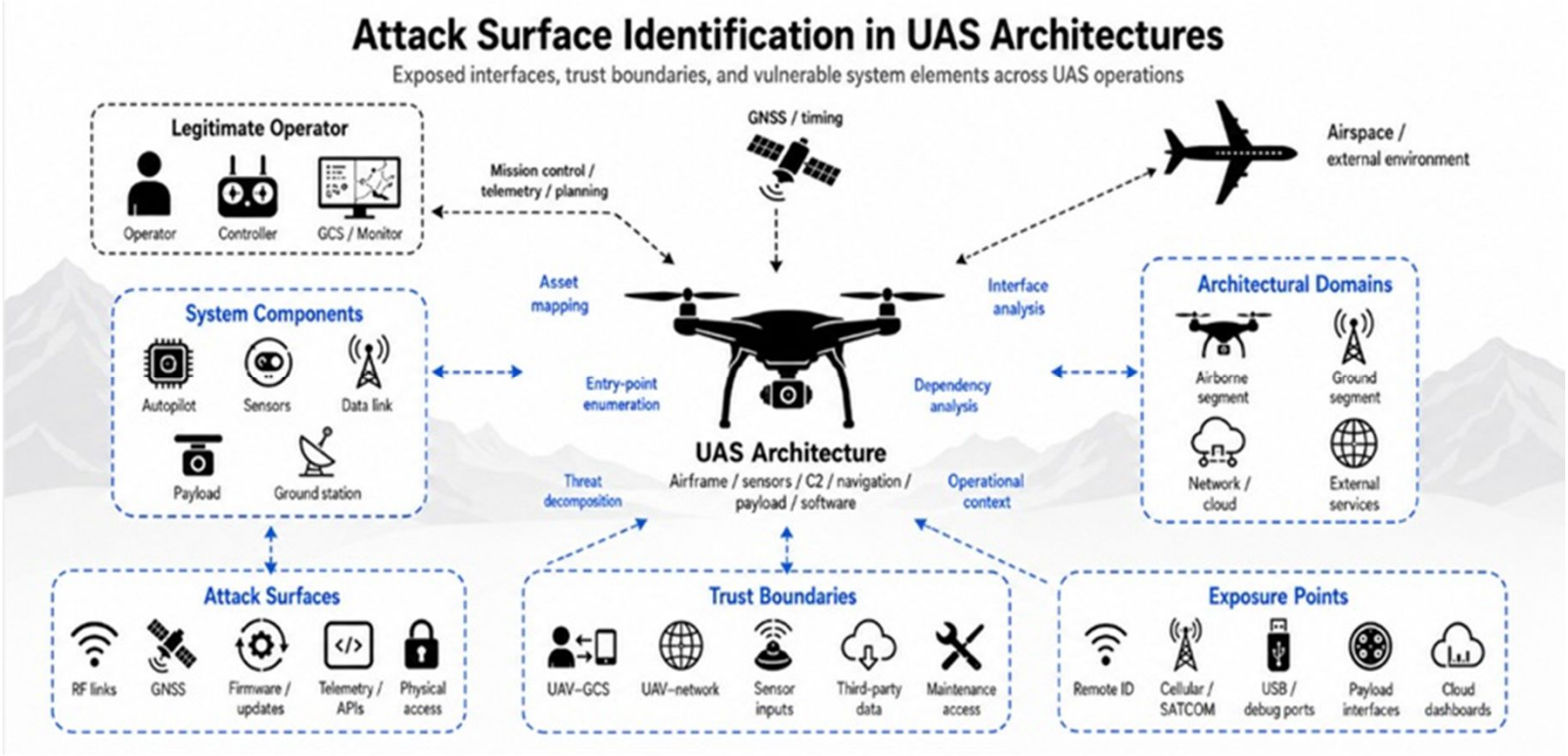


Figure S3.2. Attack surfaces are exposed interfaces. Trust boundaries mark changes in authority or trust assumptions. Inventory both before selecting test cases.

Conceptual relationships, not a demonstrated attack sequence. Teaching notes expand the assessment questions.

Threat-analysis methods and assessment outputs

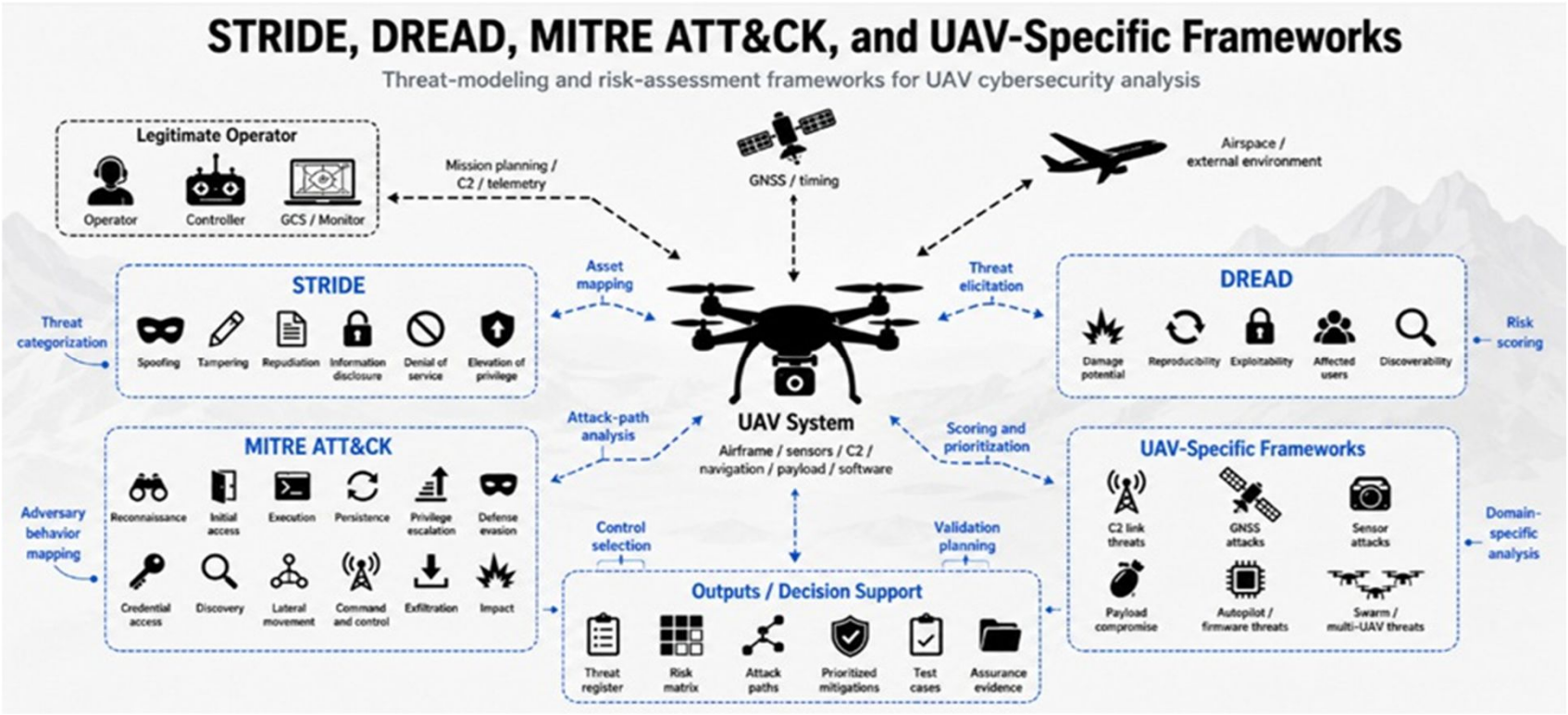


Figure S3.3. STRIDE supports threat categorization. DREAD supports relative scoring. ATT&CK describes adversary behavior. Each requires adaptation to the UAS architecture.

Conceptual relationships, not a demonstrated attack sequence. Teaching notes expand the assessment questions.

Communication, navigation, and payload assessment

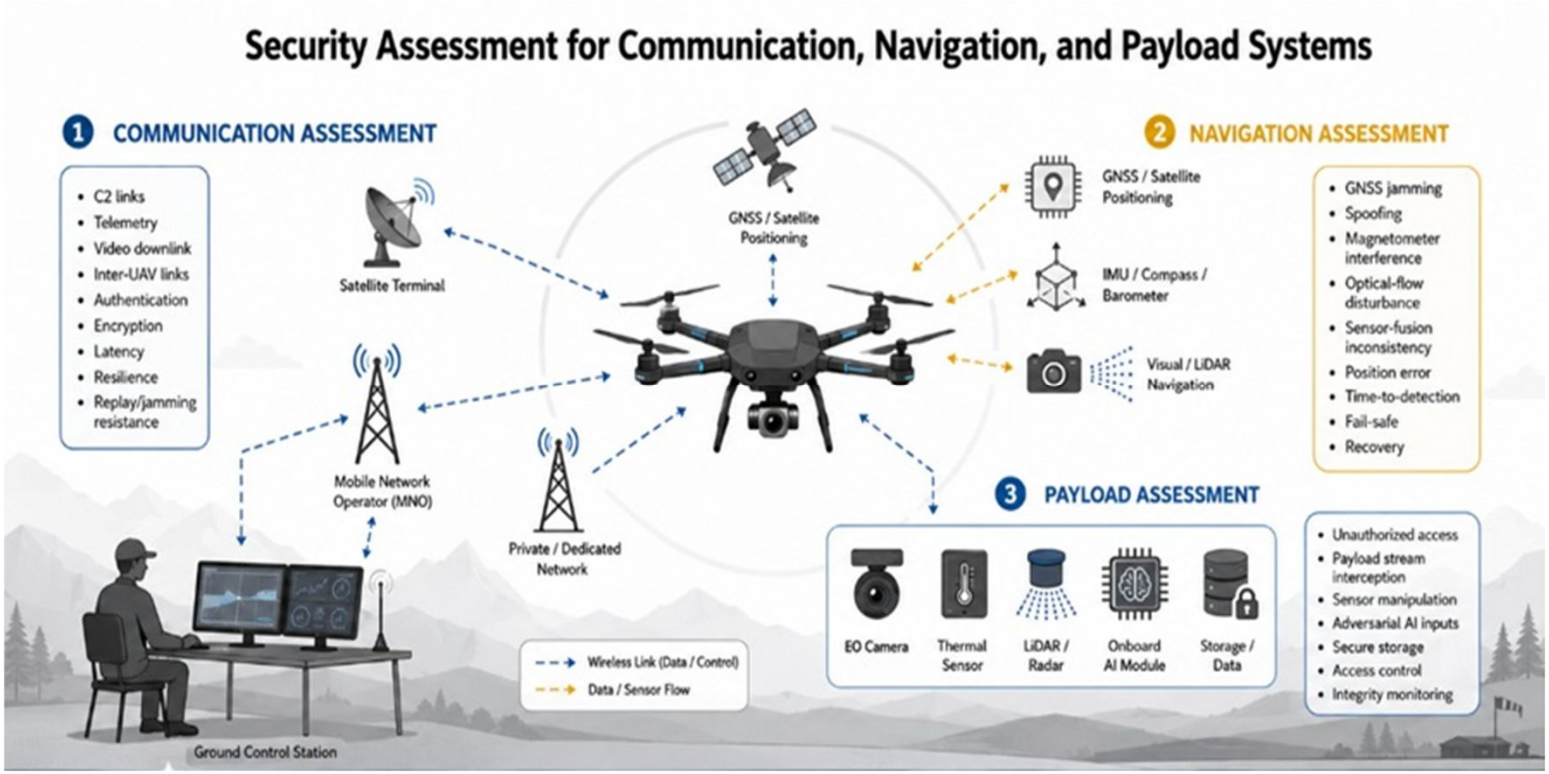


Figure S3.4. Subsystem tests should measure mission effects and recovery behavior. Dependencies can allow a failure in one subsystem to affect another.

Conceptual relationships, not a demonstrated attack sequence. Teaching notes expand the assessment questions.

Method selection and limits

The assessment question determines the method and the evidence it needs.

Method	Useful output	Limit to record
STRIDE	Threat categories applied to interfaces and data flows	Does not establish likelihood or demonstrate exploitability
DREAD	Relative prioritization using explicit scoring criteria	Subjective scores need rationale and uncertainty, not false precision
MITRE ATT&CK	Behavior descriptions and supported technique mappings	Platform coverage may leave gaps for RF and physical UAS effects
UAS-specific assessment	Mission consequences, dependencies, and testable control requirements	Document the chosen method and its validation evidence

A category, a priority score, and a successful test answer different questions.

Worked example: cooperative UWB positioning

Scope and assets	Target estimate, anchor coordinates, range records, update timing, and navigation commands.
Trust boundaries	Anchor-to-target messages, reference-position entry, dashboard access, and estimator-to-controller output.
Threat hypothesis	An incorrect or stale anchor update enters the estimator and distorts the reference geometry.
Candidate controls	Authenticated updates, freshness checks, coordinate validation, and an explicit safe pause during relocation.

Assumptions: test access is authorized, anchor references are externally established, and height constraints are known.

Evidence and acceptance criteria

Define measurable criteria before testing. Set the thresholds below for the chosen mission.

Assessment case	Evidence to collect	Example acceptance rule
Stale anchor update	Timestamp, sequence, accepted state, and estimator input trace	Reject data older than T_{max} or enter a defined degraded mode
Incorrect coordinate update	Submitted reference, validation result, reference revision, and operator log	Reject updates outside the configured operating envelope
Anchor-link interruption	Valid-range count, outage duration, mode transitions, and recovery time	Declare degraded localization within T_{detect} and apply fallback
Positioning performance	Independent ground truth, error series, validity flags, and geometry	95th-percentile error ≤ E_{max} within a declared operating domain

Course parameters: T_{max} = data-age limit, T_{detect} = detection deadline, E_{max} = permitted position error.

Practical exercise: threat register and review

Objective: produce a traceable assessment for one communication, navigation, or payload interface.

1. System definition	Identify the mission, architecture, protected assets, and assumptions.
2. Threat register	For each interface, state the threat, precondition, consequence, and current control.
3. Priority and confidence	Explain likelihood and impact separately. Record evidence gaps and uncertainty.
4. Validation and decision	Define a test, acceptance criterion, owner, and residual-risk review.

Deliverable: one annotated architecture, a threat register, and a short evidence-based review.

References and terminology

Primary references support the method descriptions and operational-context notes.

Microsoft Threat Modeling for Drivers. STRIDE categories and DREAD approach.
learn.microsoft.com/en-us/windows-hardware/drivers/driversecurity/threat-modeling-for-drivers

MITRE ATT&CK FAQ. Tactics, techniques, and use of the knowledge base.
attack.mitre.org/resources/faq/

NIST SP 800-30 Rev. 1, Guide for Conducting Risk Assessments (2012).
csrc.nist.gov/pubs/sp/800/30/r1/final

EASA Explanatory Note to ED Decision 2025/018/R (2025).
SORA 2.5 terminology: BVLOS with airspace observer.

UAS: unmanned aircraft system GCS: ground control station C2: command and control
PNT: positioning, navigation, and timing UWB: ultra-wideband